

ICU EDGE GATEWAY

A verified reference implementation for real-time ICU vital-signs ingestion, deterministic NEWS2 clinical risk scoring, and FHIR R4 interoperability.

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THE PROBLEM

ICU bedside monitors generate continuous, high-frequency vital-signs data that rarely reaches the EMR in a structured, standards-compliant form. Nurses are exposed to alarms triggered by sensor noise and disconnected leads rather than true deterioration — a documented driver of alarm fatigue. Deterioration scoring, when it exists, is frequently manual, delayed, or inconsistent across shifts and does not distinguish clinically important sub-populations such as chronic hypercapnic respiratory failure patients, who require a different SpO₂ reference scale.

WHAT THE GATEWAY DOES

- Real-time HL7 ingestion.** Asynchronous HL7 v2.5.1 ORU^R01 parsing over MLLP (VT/FS/CR framing, AA/AE acknowledgement, malformed messages NAK'd without dropping the connection); multi-vendor normalization across proprietary OBX codes with no per-vendor configuration required.
- Artifact rejection.** Physiological bounds checking suppresses the score rather than fabricating one — an out-of-bounds reading is marked REJECTED and produces NO SCORE, not a reassuring zero.
- Deterministic NEWS2 scoring.** All seven RCP 2017 parameters, every scoring boundary tested at the exact threshold, including the non-monotonic 41–50 bpm band, both SpO₂ reference scales (materially changing outcome for hypercapnic patients), and AVPU/new-confusion routing.
- FHIR R4 output.** Bundle and Observation resources with dual LOINC + SNOMED coding and UCUM units, a NEWS2 Observation coded SNOMED CT 1239842005, and seven scored components — structured for direct EMR ingestion without custom mapping work.
- Advisory-only posture enforced in the transport.** Not only documented: an X-CDS-Advisory-Only: true response header, a FHIR extension, and a Bundle note accompany every response.

EVIDENCE, NOT ADJECTIVES

816 / 1	94.6%	0 → 3 → 6 → 7 → 17	6 phases
automated tests passing / xfailed (CI-confirmed)	statement coverage (CI-measured)	NEWS2 sepsis-scenario deterioration trace, live	zero-regression discipline across engineering phases

SpO₂ scale selection alone moves an identical patient from **2 LOW** (Scale 2) to **5 MEDIUM** (Scale 1) — a 3-point swing across a clinical risk band, verified live over HTTP. 42 dedicated non-finite-value (NaN/±Inf) rejection tests, 18 of which fail against the pre-fix code — proof the tests are load-bearing, not decorative.

REGULATORY POSTURE — STATED PLAINLY

The gateway is engineered under IEC 62304 Class B software life-cycle practices with ISO 14971 risk analysis applied to each hazard, and is designed against the non-device clinical decision support criteria in FD&C Act §520(o)(1)(E) as interpreted by FDA's Clinical Decision Support Software guidance. Output is advisory only and requires independent clinician review; the software triggers no automated treatment. These are the developer's own determinations and have not been reviewed by FDA or any notified body.

WHAT THIS IS — AND WHAT IT ISN'T

THIS IS	THIS IS NOT
A verified reference implementation and IP asset: the HL7 → artifact rejection → NEWS2 → FHIR R4 pipeline, built to medical-software engineering discipline, with the evidence trail a regulatory reviewer would ask for.	A production-ready, FDA-cleared, or clinically validated device. No real patient data has been processed; no deployment, load test, or security hardening (TLS, authn/z, CORS lock-down) has occurred; no regulatory body has reviewed it.

ENGAGEMENT MODELS

Available for IP / source-code licensing or acquisition, per-bed or per-site subscription licensing to an existing compliant healthtech vendor or EMR integrator, or an asset/acqui-hire transaction. Full structure and indicative pricing in the accompanying *B2B IP Licensing & Pricing Framework*.

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